SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	29 June2026
Team ID	
Project Name	Rising Waters: Smart AI Flood Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML5, CSS3, JavaScript, Bootstrap 5	*Provides a responsive, user-friendly interface for entering weather parameters and displaying flood prediction results. Bootstrap ensures mobile-friendly design.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python 3.11, Flask*	*Flask is a lightweight web framework used to handle user requests, process prediction inputs, and communicate with the machine learning model efficiently.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	***CSV Dataset, Joblib (Model Storage)*** *(or SQLite if implemented)*	*The historical weather dataset is stored in CSV format for training. The trained machine learning model is saved using Joblib for efficient loading during prediction. SQLite can optionally be used to store prediction history.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*IBM Cloud*	*IBM Cloud provides a reliable and scalable platform for deploying the Flask application, allowing users to access the flood prediction system from anywhere with high availability.*
5	**Version Control & CI/CD**	*Git*	*Git and GitHub are used for source code management, version control, team collaboration, tracking project changes, and maintaining backups of the application.*

	(e.g., GitHub, GitLab, Jenkins)	*, GitHub*	
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib, Jupyter Notebook, IBM Cloud*	***Scikit-learn and XGBoost*** *are used for training and evaluating flood prediction models.* ***Pandas*** *and* ***NumPy*** *handle data preprocessing and numerical computations.* ***Matplotlib*** *and* ***Seaborn*** *are used for data visualization and exploratory analysis.* ***Joblib*** *saves and loads the trained model for deployment.* ***Jupyter Notebook*** *is used for model development and experimentation, while* ***IBM Cloud*** *hosts the deployed Flask application for remote access.*